Dominic Catania

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Education

- **Texas A&M University** — College Station, TX
  B.S. Computer Engineering; University Honors; GPA: 3.73 — *Aug 2024 – May 2028*
  - Data Structures & Algorithms; Computer Systems & Architecture; Digital System Design; Probability & Statistics

Skills

- **Languages**: C, C++, Python, Java, R, SQL, HTML, CSS, Javascript, Typescript
- **Technologies**: Git, Linux, Arduino, Pandas, Numpy, Tensorflow, Pytorch, Scikit-learn, Matlab, Docker, Kubernetes, AWS S3, Gitlab, Bash, Next.js, Node.js, REST API, GraphQL, Tableau, Power BI, SQLite, MySQL, PostgreSQL, Django, React, Bootstrap, Agile Development, Jira, Microsoft Office, Google Suite

Professional Experience

- **Depth Sensor and Turtle Tracker** — May 2026 - Present
  Tortugas Costa Rica - Sea Turtle Conservation Research
  - Drove development of embedded sea turtle telemetry system from initial concept to functional prototype, architecting a C++ ESP32-based system for autonomous marine deployments with reliable low-power operation
  - Integrated modular software and hardware supporting pressure sensing, RTC synchronization, SD storage, environmental monitoring, and Iridium satellite communications to enable autonomous field data collection
- **Vigil - Data Platform** — June 2024 - May 2025
  FIRST Robotics Competition Team 4206
  - Designed and implemented a full-stack analytics platform with interactive Python Plotly visualizations for robotics performance evaluation and anomaly detection, supporting real-time competition decisions
  - Integrated Ollama AI-assisted analysis to identify performance regressions and optimize match strategy
- **Carson - Competition Robot** — Jan 2024 – May 2024
  FIRST Robotics Competition Team 4206
  - Developed autonomous software stack using Java and C++ integrating AprilTag localization, vision-assisted targeting, distance interpolation, and autonomous path planning for high-speed competition
  - Designed driver-assistance features including automatic alignment, automated game piece acquisition, and assisted scoring by coordinating vision, localization, and motion-control subsystems

Leadership Experience

- **Triangle Fraternity** — Oct 2024 - Present
  Vice President of External Relations
  - Coordinated partnerships across engineering organizations and industry representatives
  - Managed cross-functional initiatives spanning recruitment, public relations, and chapter operations
- **FIRST Robotics Competition Team 4206** — Aug 2021 - May 2024
  Team Captain; Driver; Lead Programmer
  - Led a 30+ member robotics team, overseeing project management, system integration, and collaboration
  - Mentored 10 new members in sensor integration, computer vision, and autonomous systems enhancing robot precision

Awards

- **2025 TAMU Datathon Second Place**
- **2024 FIRST Robotics Championship World Semifinalists**
- **Reta & HJ Haynes University Scholarship**